signal and noise

prof. dr. nicolas meseth

1. what the numbers say
2. distance against spread
3. build evidence

part 1

what the numbers say

the lamp is as bright as ever, so why does its beam end so soon?

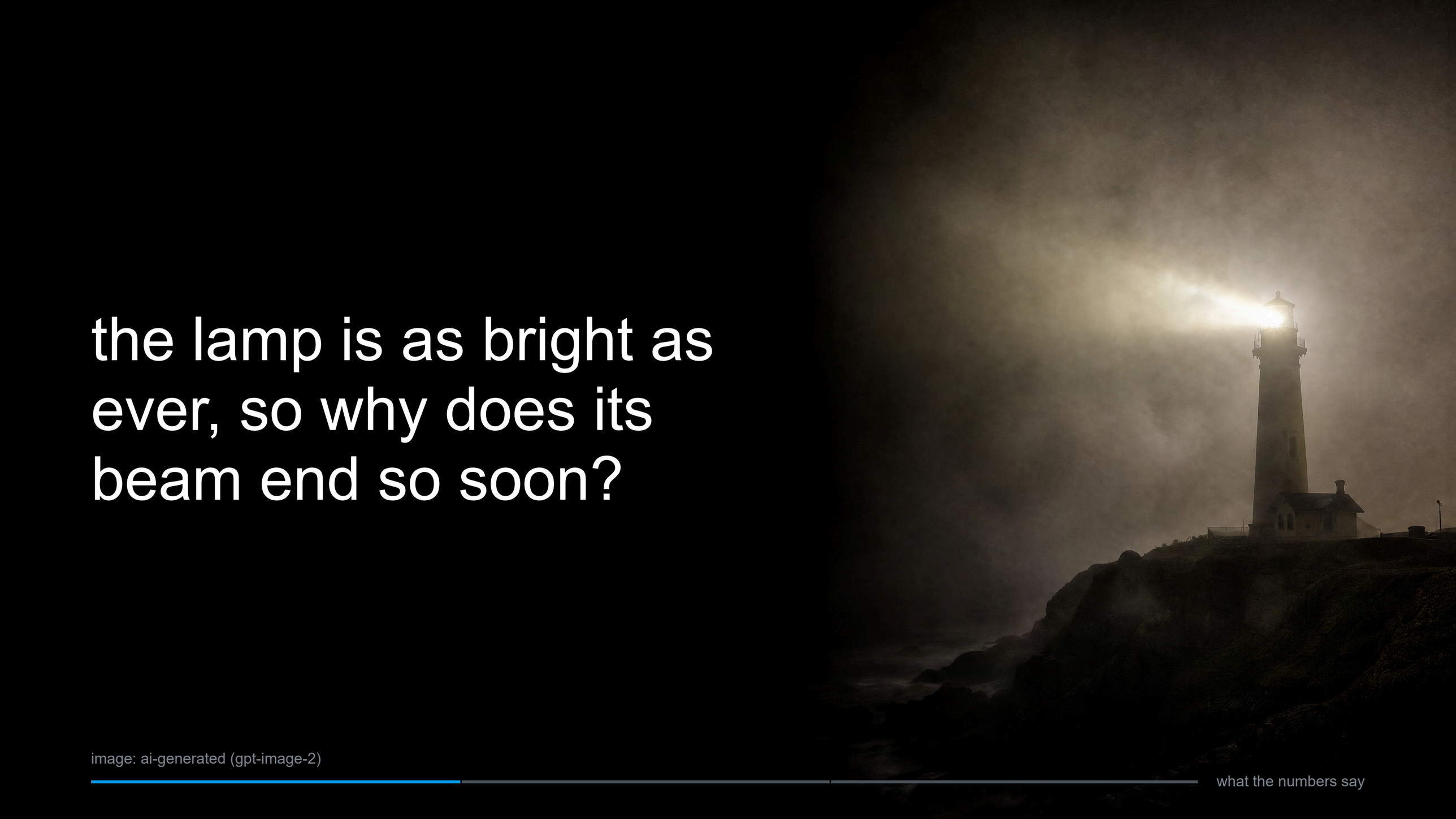
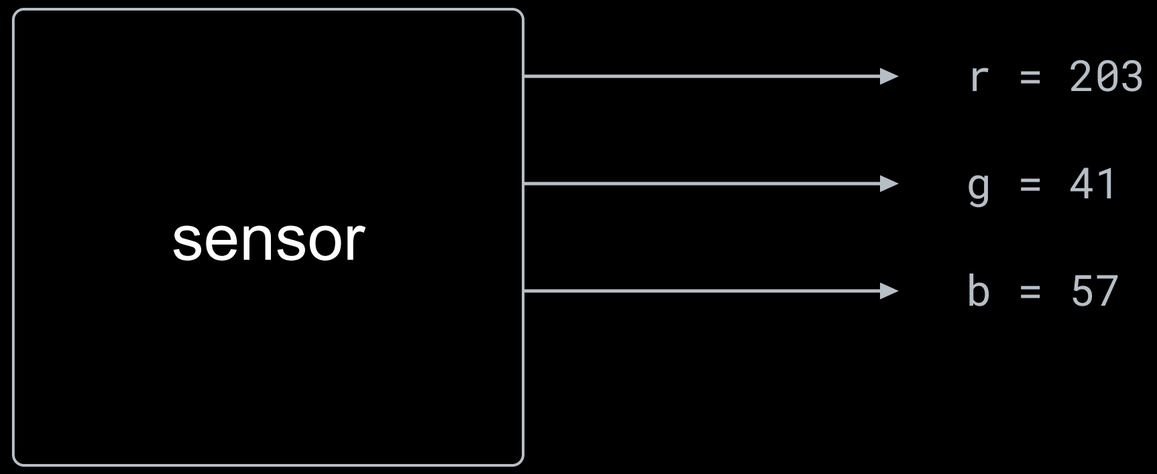


image: ai-generated (gpt-image-2)

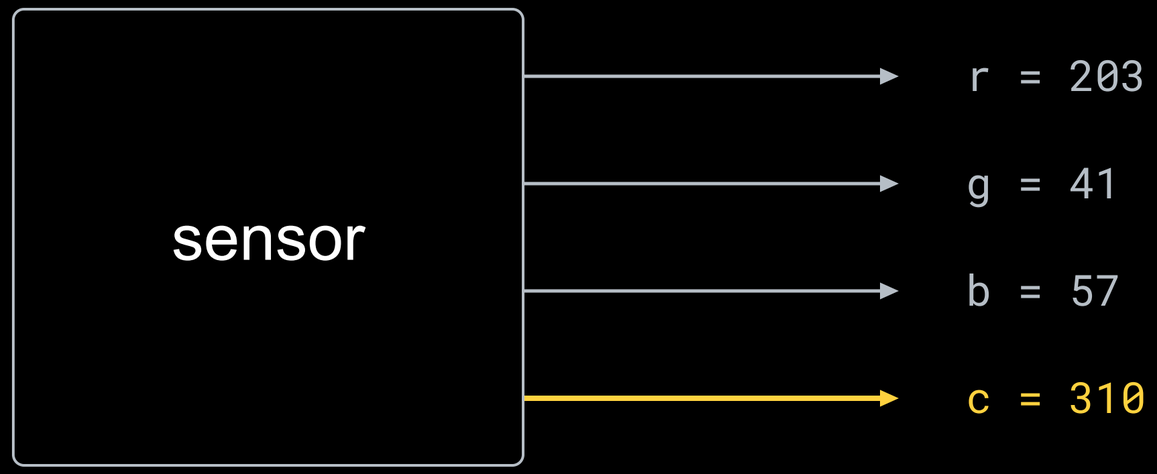
how many colours does your alphabet have?

every team works with the same LED and sensor, and yet every team did something different.

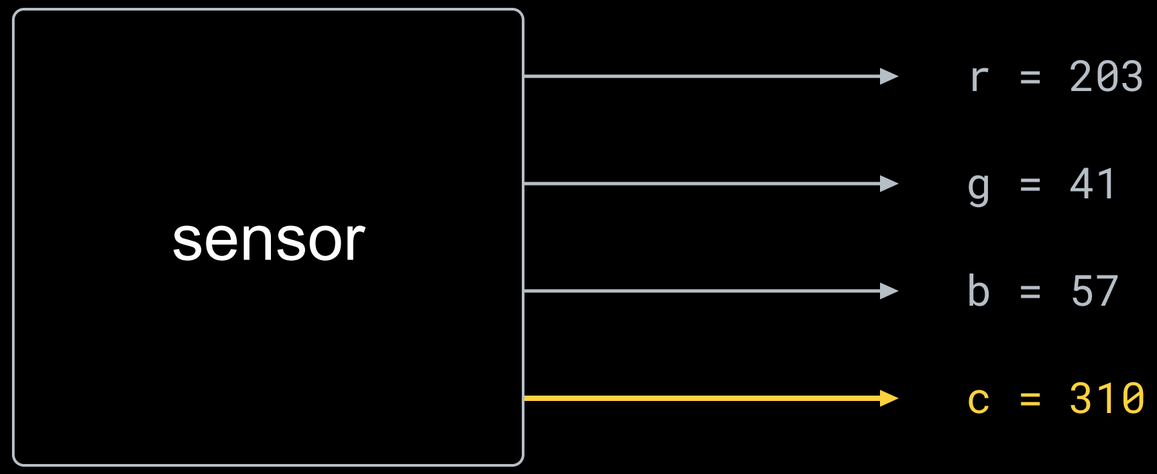
the sensor returns four numbers



the sensor returns four numbers



the sensor returns four numbers



`clear`

no colour filter

most sensitive

your eyes see a colour, but the program only sees four numbers.

sent is not measured

what you send

```
r = 255
g =   0
b =   0
```

sent is not measured

what you send

```
r = 255
g =   0
b =   0
```


what arrives

```
r = 203
g =  41
b =  57
c = 310
```

the filters overlap, distance dims the light, and the room shines in too

sent is not measured

what you send

```
r = 255
g =   0
b =   0
```


what arrives

```
r = 203
g =  41
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c = 310
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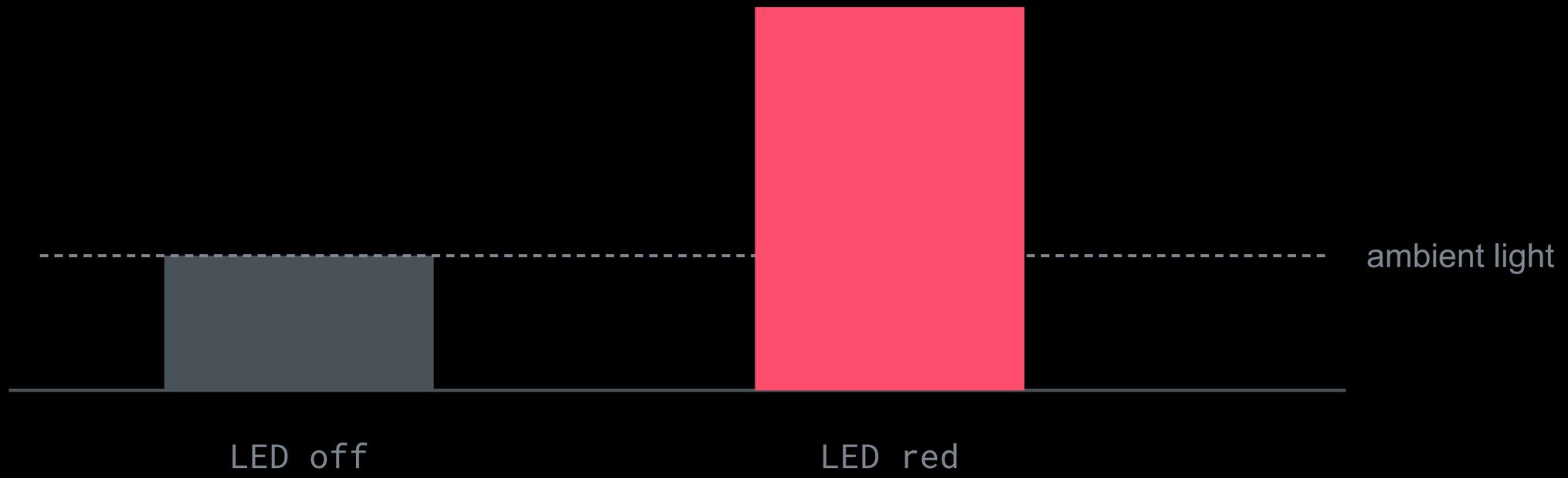
the filters overlap, distance dims the light, and the room shines in too

the numbers differ from what was sent, and that is normal, not a defect.

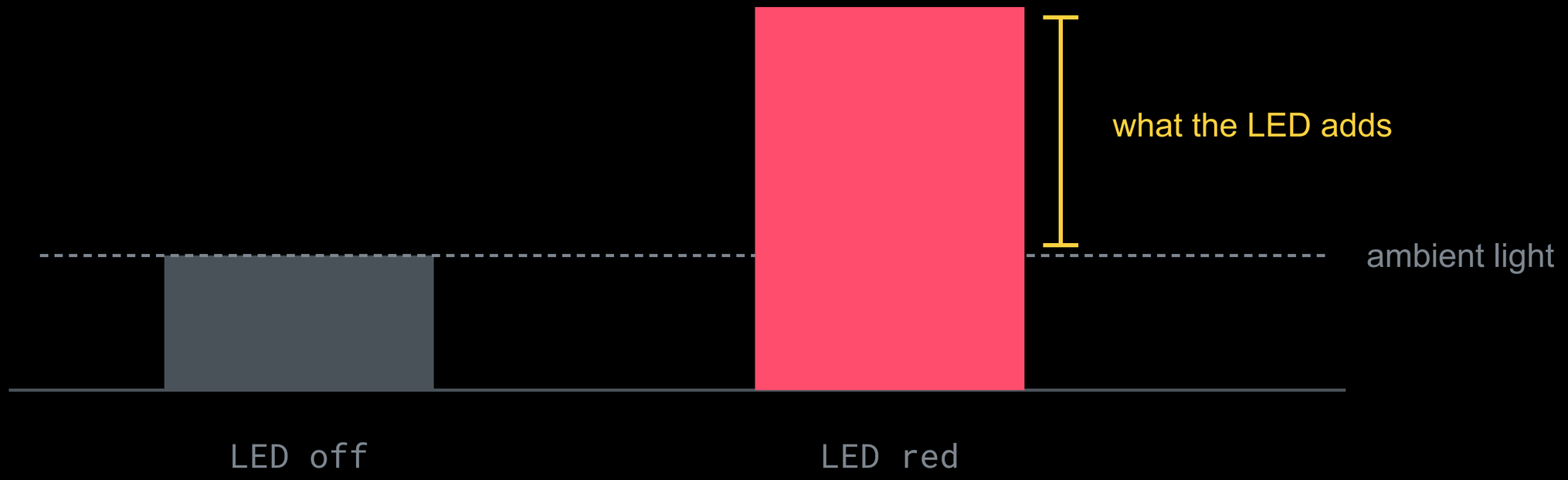
measure the dark first



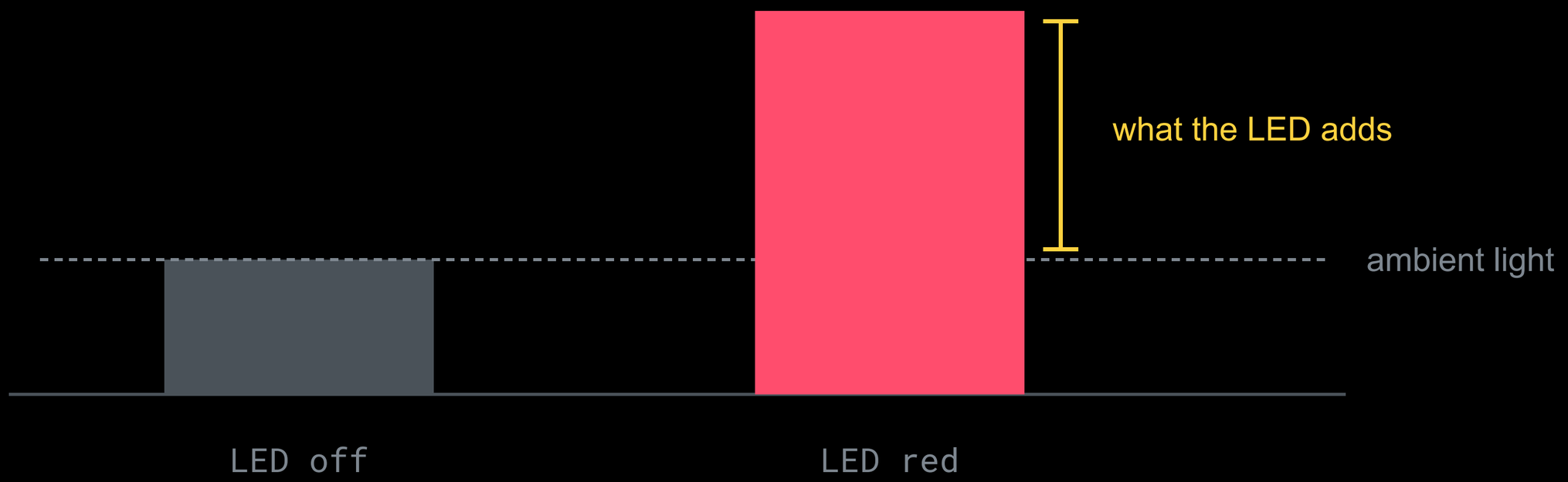
measure the dark first



measure the dark first

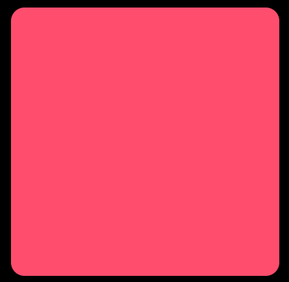


measure the dark first




```
reading = ambient + LED + noise
```

ten readings of one colour

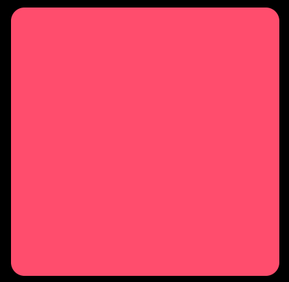


LED red
ten readings

what the sensor reports

214

ten readings of one colour

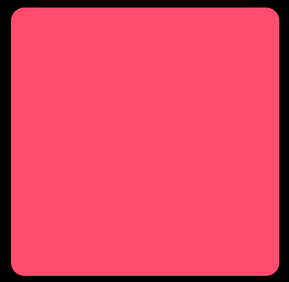


LED red
ten readings

what the sensor reports

214 211 218 209 216 213 220 207 215 212

ten readings of one colour



LED red
ten readings

what the sensor reports

214 211 218 209 216 213 220 207 215 212

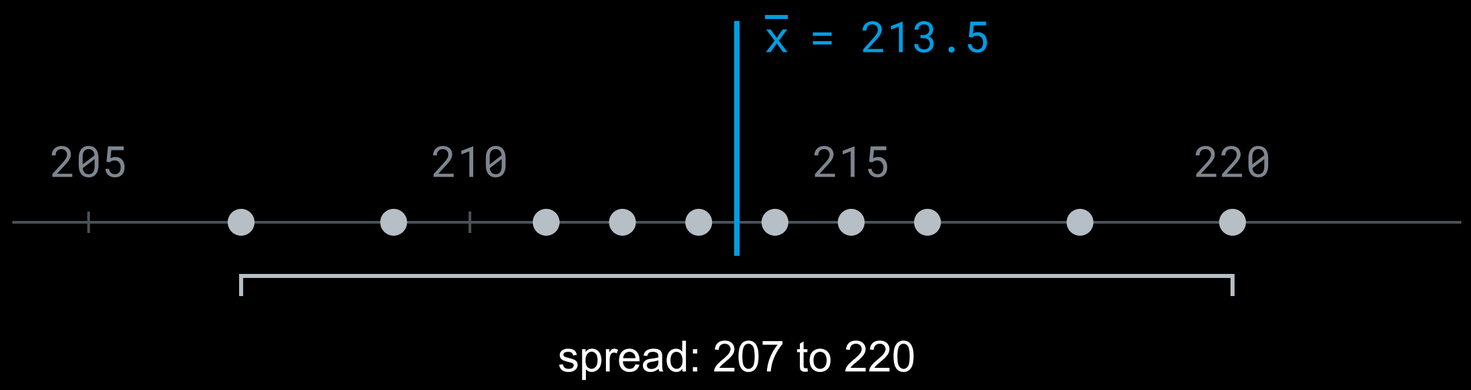


spread: 207 to 220

LED red
ten readings

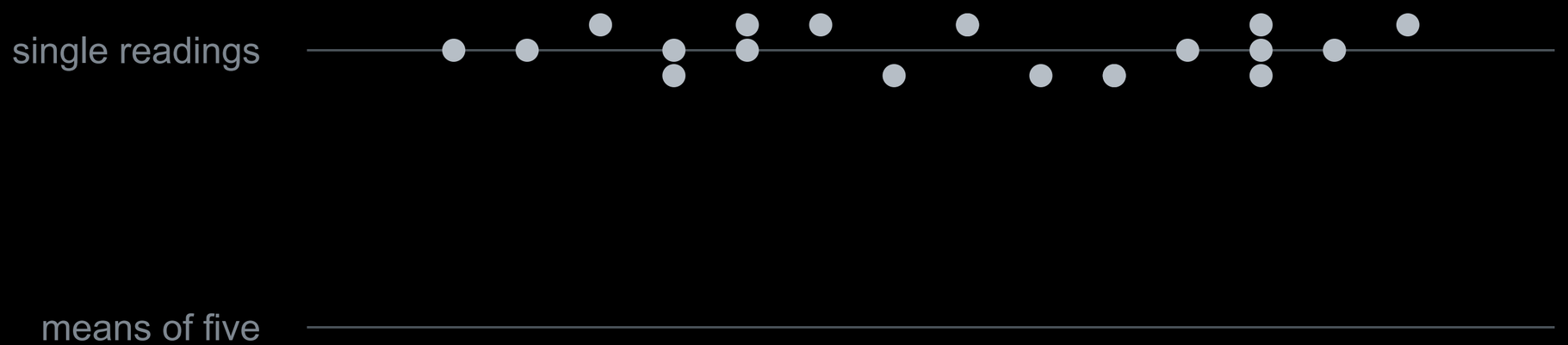
what the sensor reports

214 211 218 209 216 213 220 207 215 212

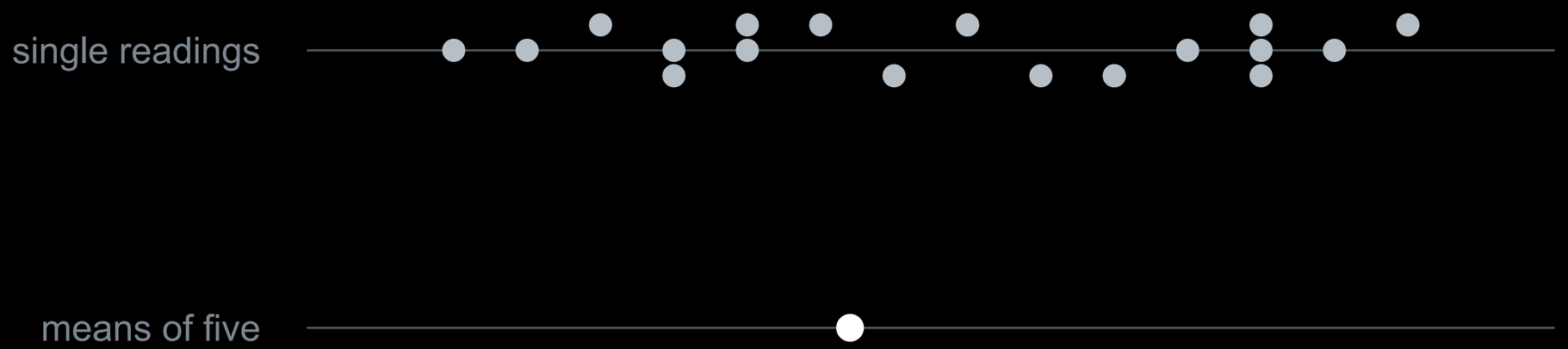


nothing changed between the ten readings, and still every number is different.

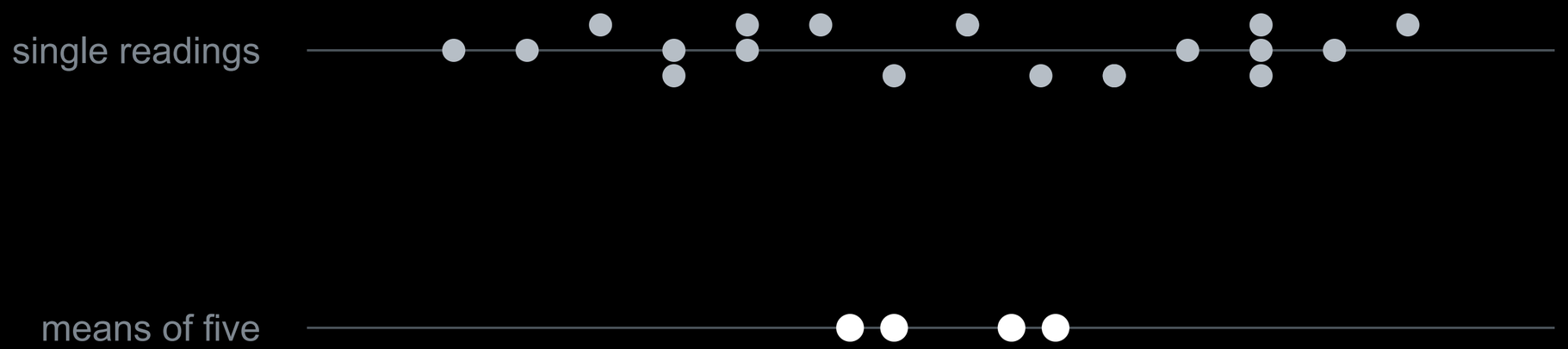
five readings, one mean



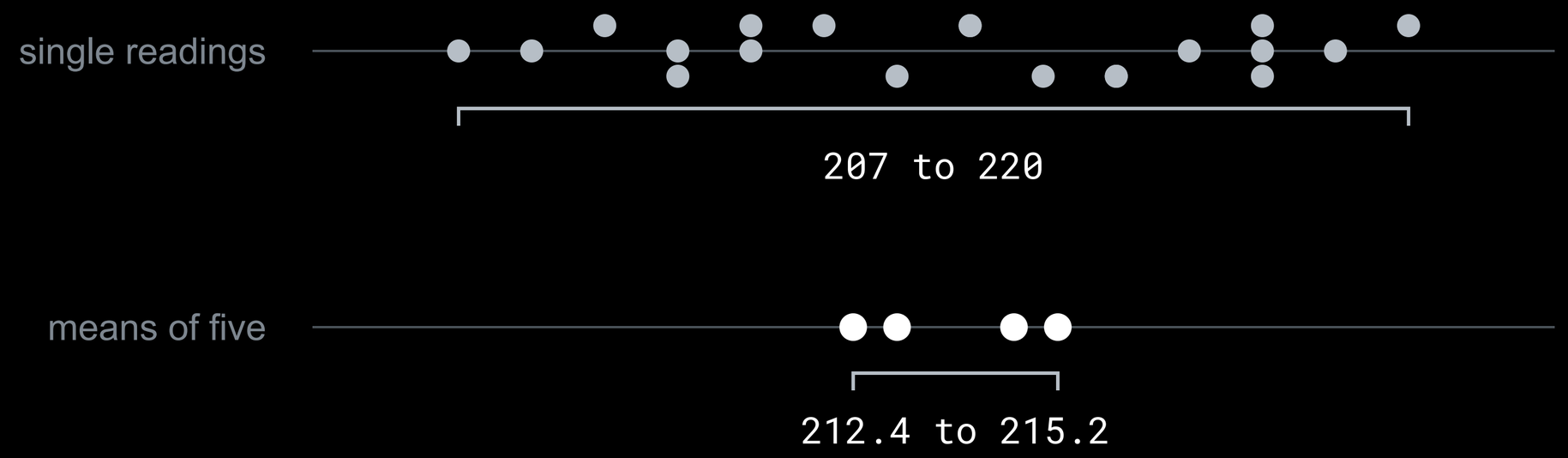
five readings, one mean



five readings, one mean



five readings, one mean



a mean of five scatters less than a single reading, but you get one value instead of five.

the scatter decides how many colours your alphabet can have.

your code has to take it into account, so measure how far your readings move.

part 2

distance against spread

the same signal, averaged over longer windows

raw signal  288 readings

distance against spread

the same signal, averaged over longer windows

raw signal  288 readings

mean of 4  72 means

the same signal, averaged over longer windows

raw signal



288 readings

mean of 4



72 means

mean of 12



24 means

the same signal, averaged over longer windows

raw signal



288 readings

mean of 4



72 means

mean of 12



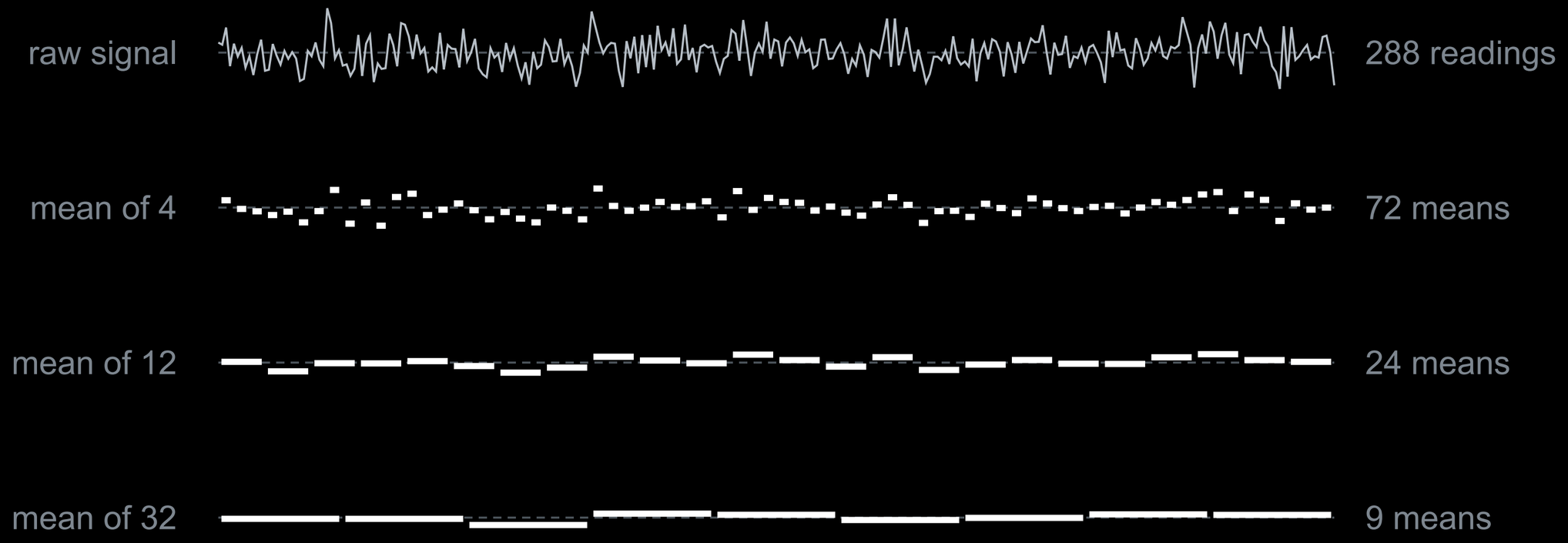
24 means

mean of 32



9 means

the same signal, averaged over longer windows

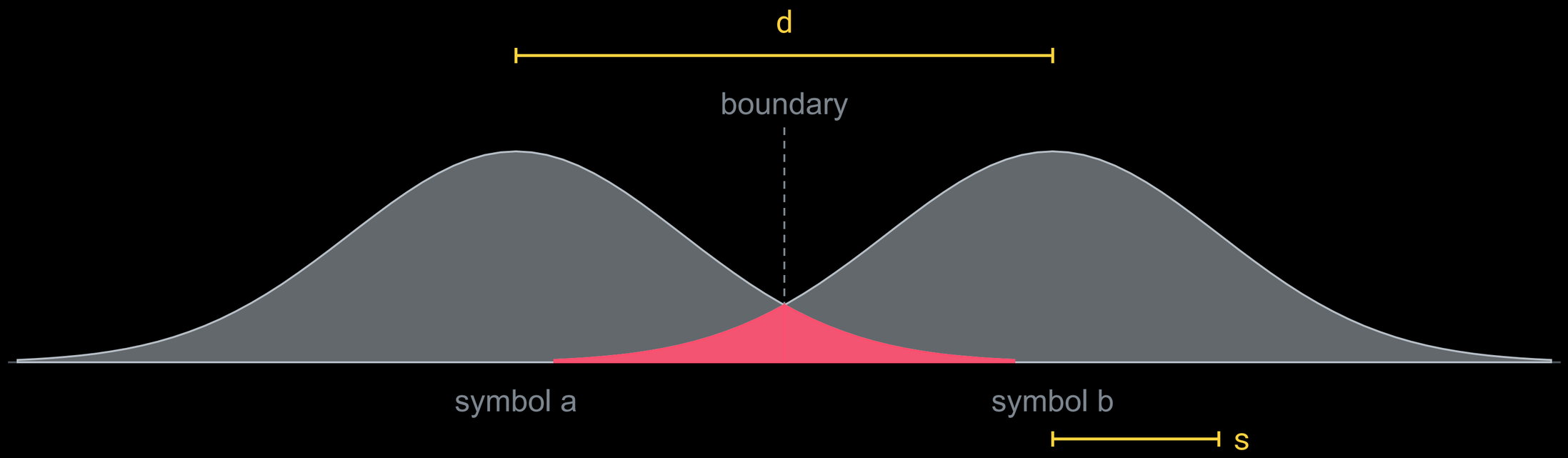


longer windows give calmer values, but fewer of them per second.

the noisy sensor: hands-on-computer-science.de/cs/en/items/noisy-sensor

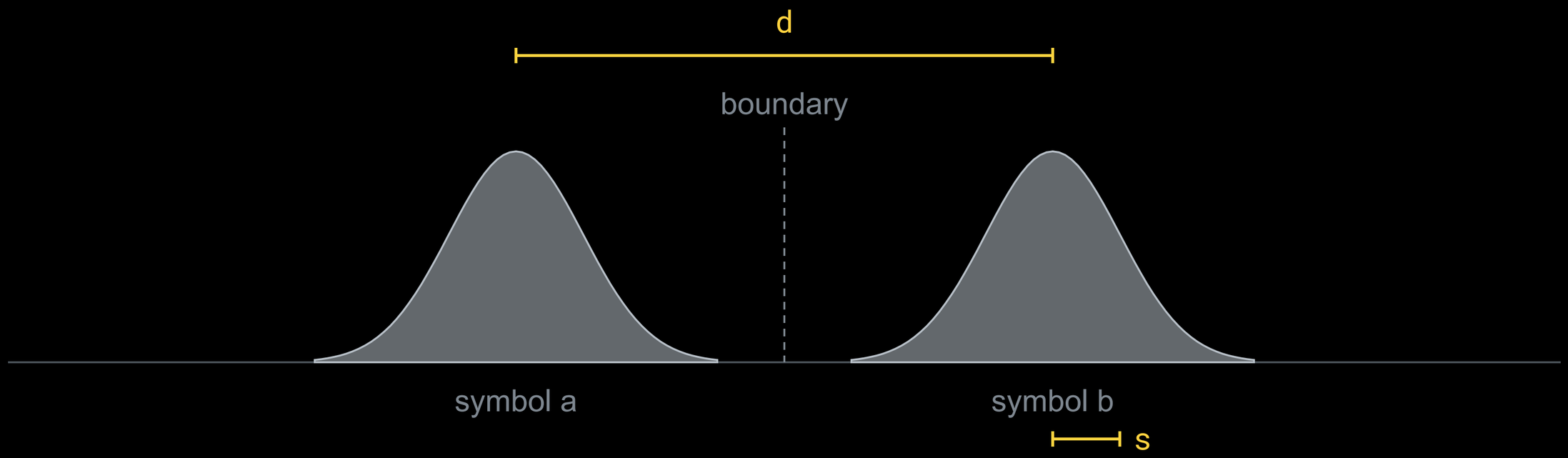
send 25 symbols: which lane decides wrongly, and how often?
turn the noise up: which lane still gets every symbol right?
what does the mean of ten cost in symbols per second?

two symbols, two clouds



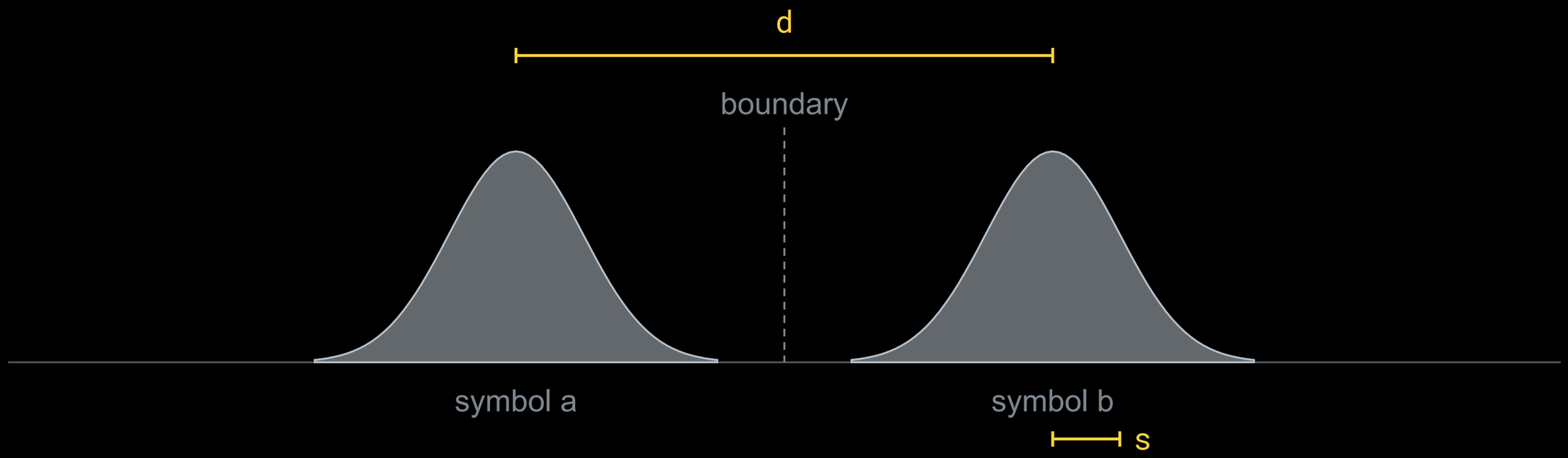
some readings land on the wrong side

two symbols, two clouds



every reading lands on its own side

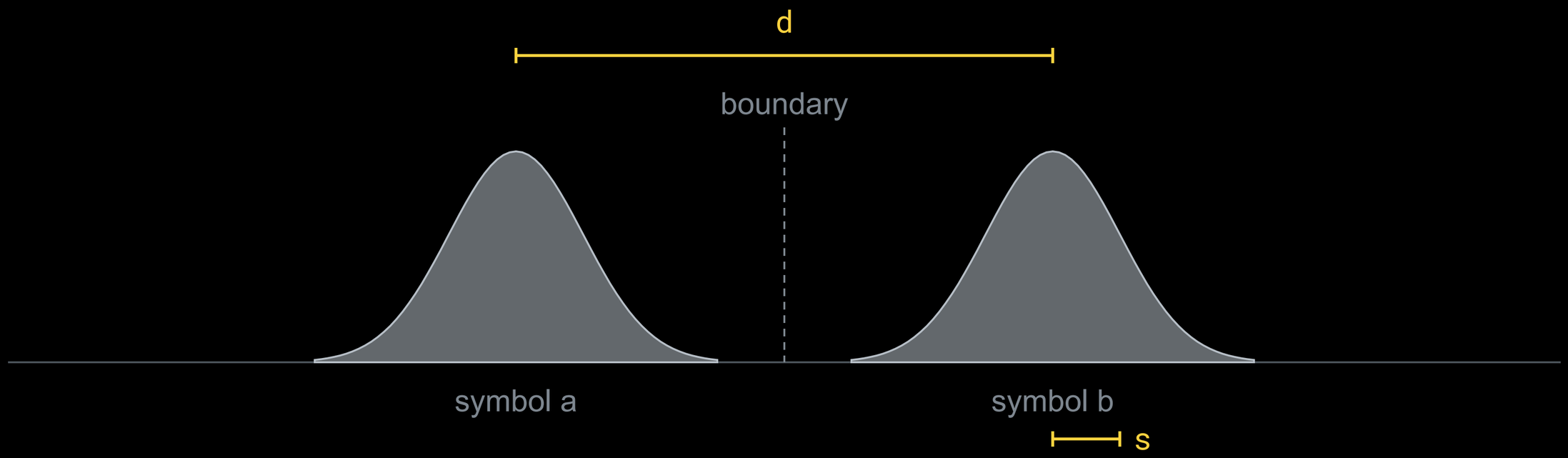
two symbols, two clouds



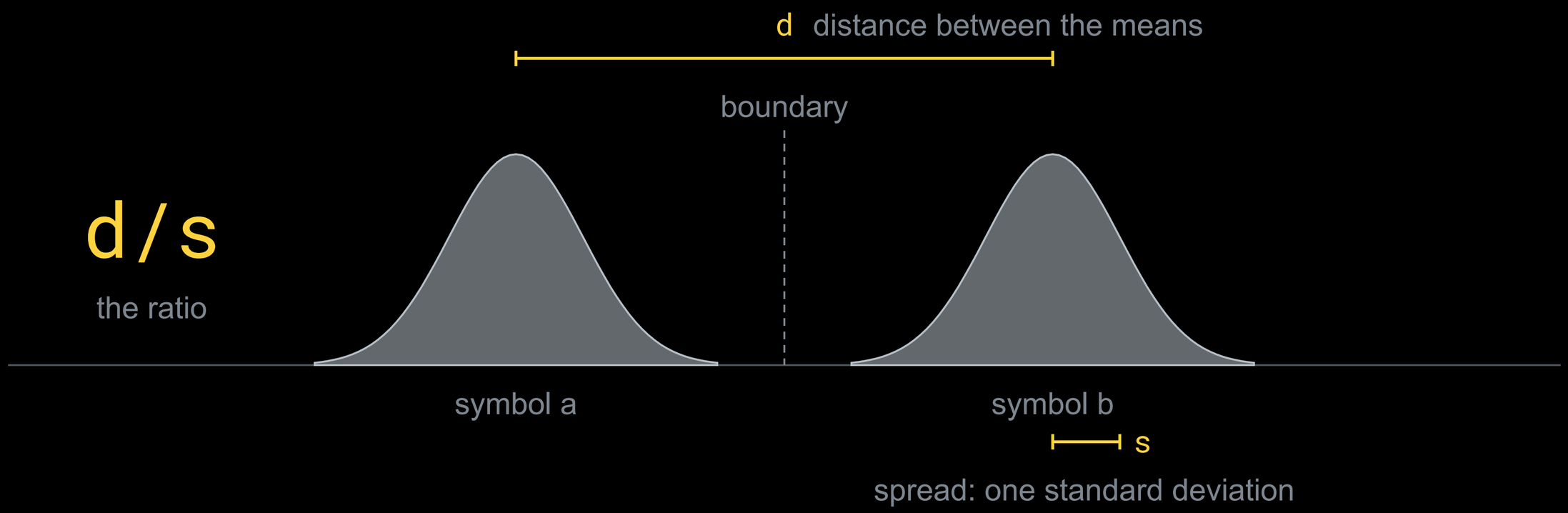
every reading lands on its own side

at the same distance, narrower clouds keep every reading on its own side.

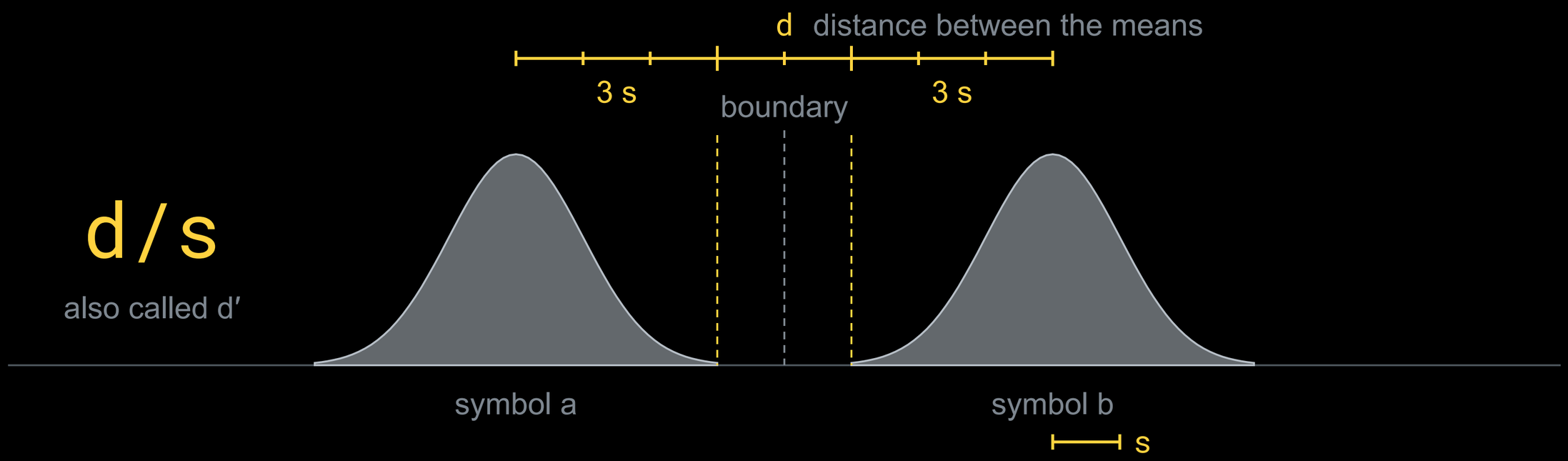
the ratio is what counts



every reading lands on its own side



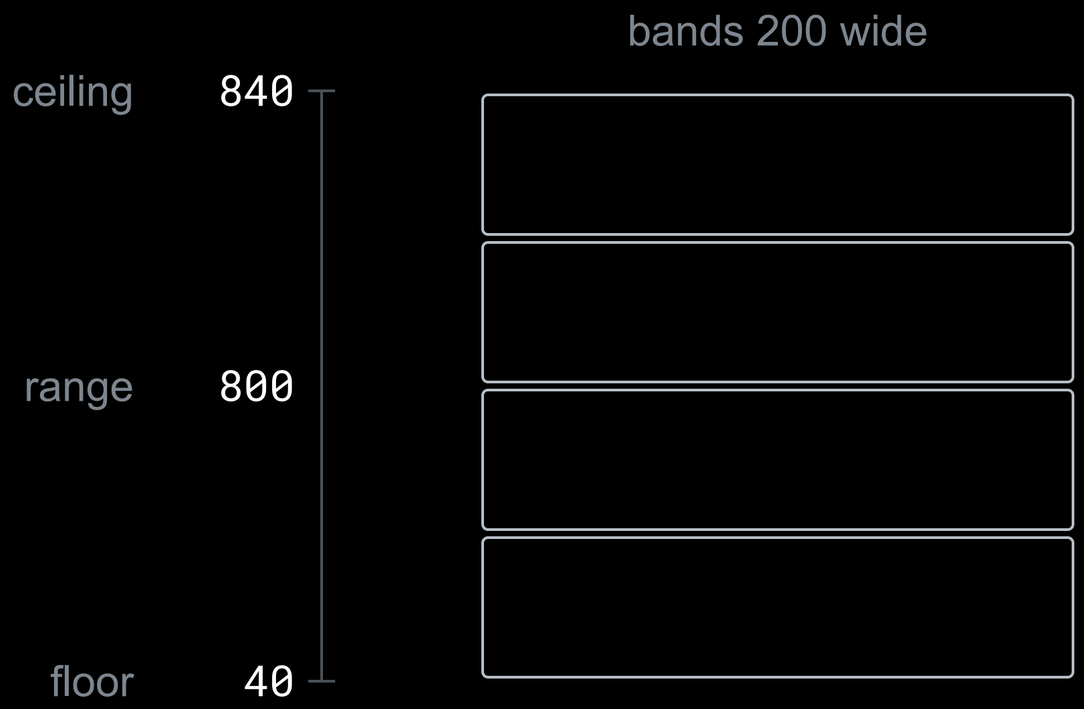
d/s counts how many spreads fit between the two means



each cloud reaches 3 s, so above d/s = 6 they stay apart; here d/s = 8

your alphabet size measures nothing but this ratio.

how many symbols fit?



800 / 200 = 4

how many symbols fit?

ceiling 840

range 800

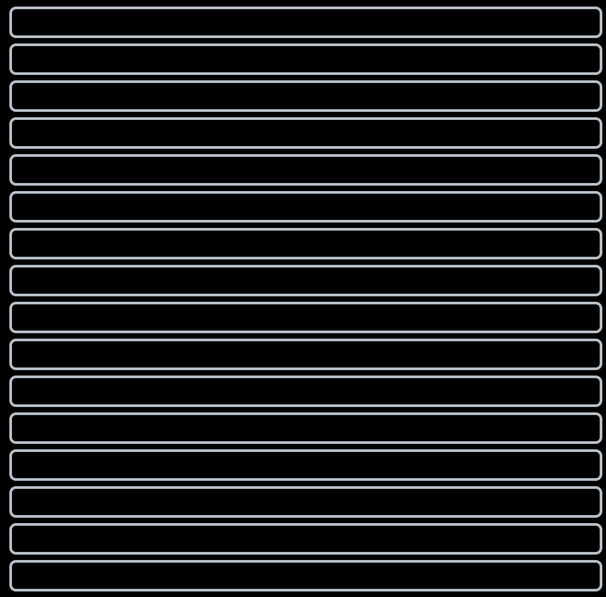
floor 40

bands 200 wide



800 / 200 = 4

bands 50 wide



800 / 50 = 16

how many symbols fit?

ceiling 840

range 800

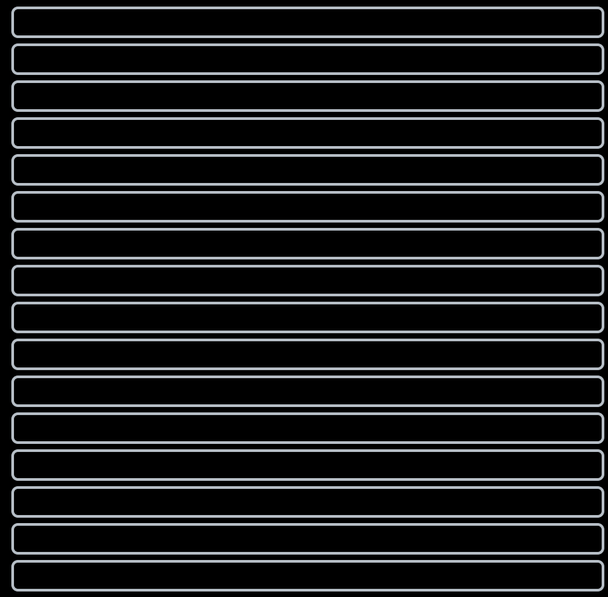
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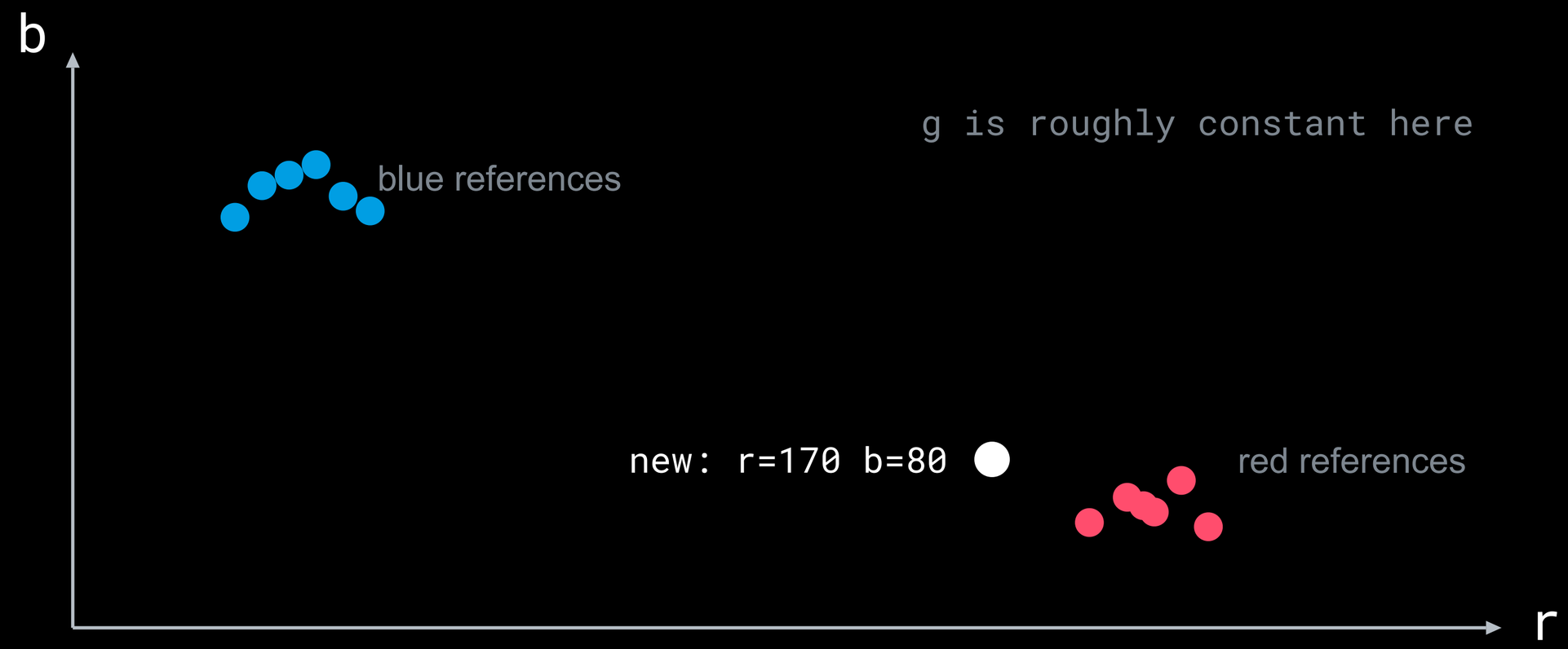
the range divided by the width of one band gives the size of your alphabet.

calibrate, then classify



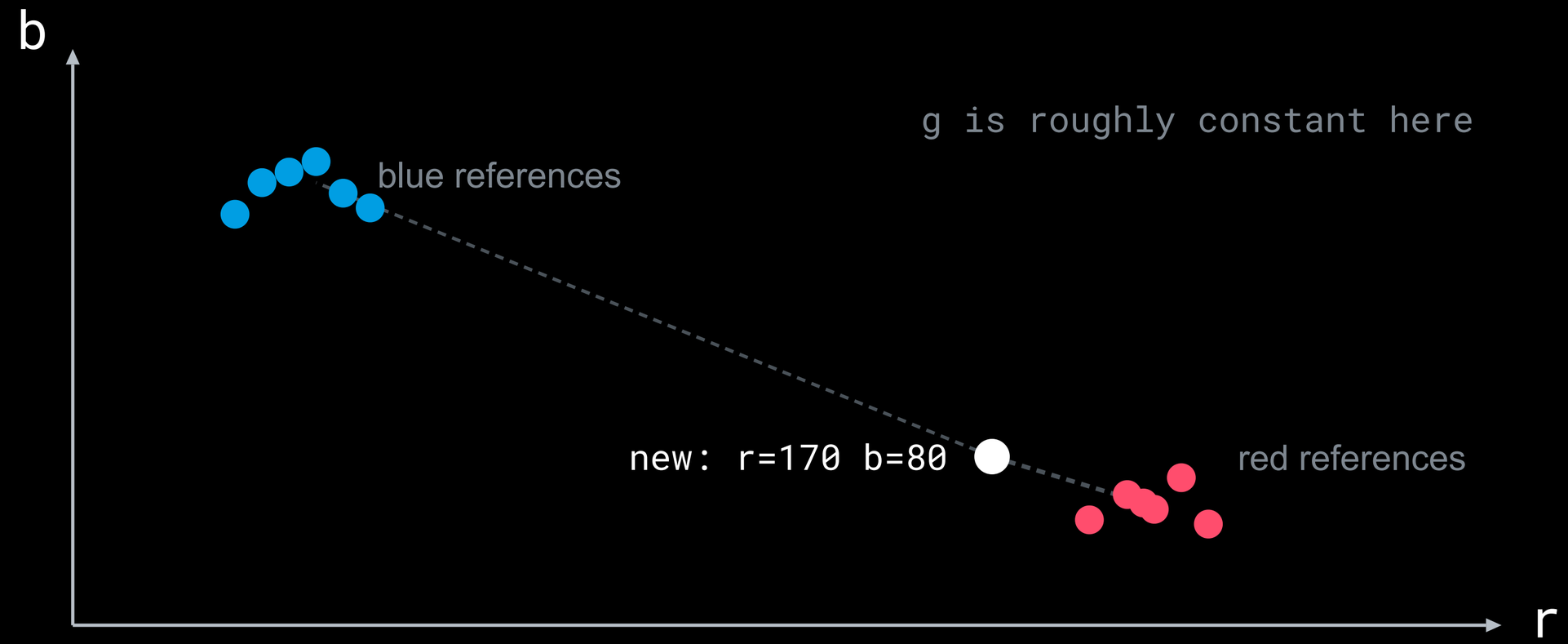
measure known colours many times

calibrate, then classify



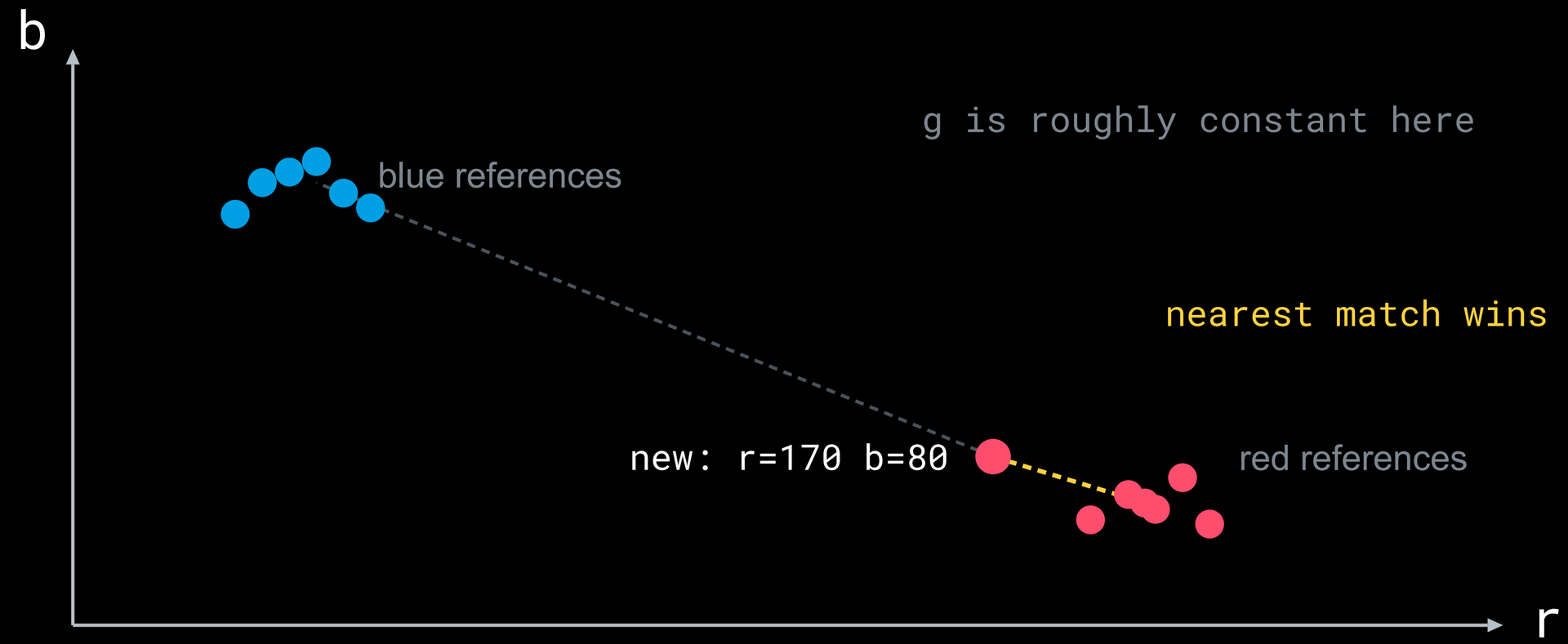
measure known colours many times

calibrate, then classify



measure known colours many times

calibrate, then classify



measure known colours many times

eight symbols, two window lengths

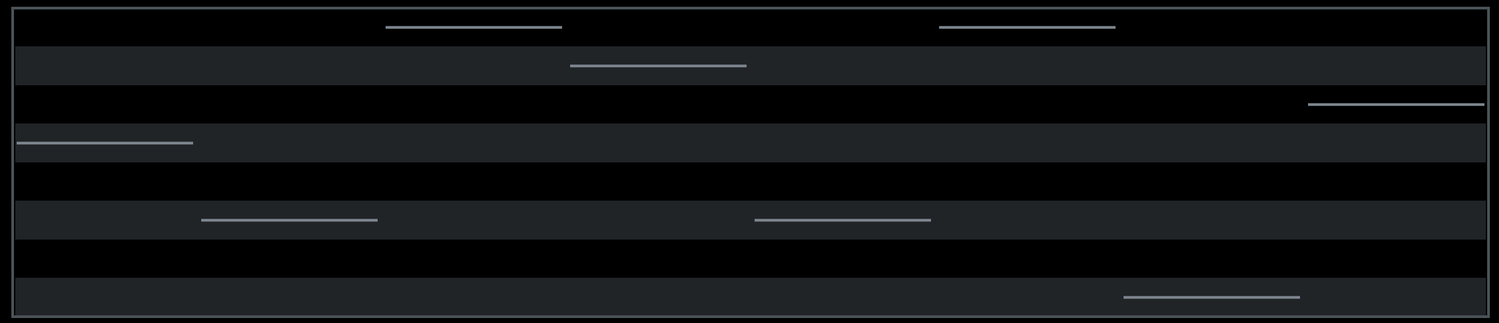
short window, 25 ms



6 of 32 wrong

40 symbols/s

window four times as long, 100 ms



the same 0.8 s in both tracks; each bar is one reading, red in a wrong band

simulated readings; the trade-off is the one your own link shows

distance against spread

eight symbols, two window lengths

short window, 25 ms



6 of 32 wrong

40 symbols/s

window four times as long, 100 ms



0 of 8 wrong

10 symbols/s

the same 0.8 s in both tracks; each bar is one reading, red in a wrong band

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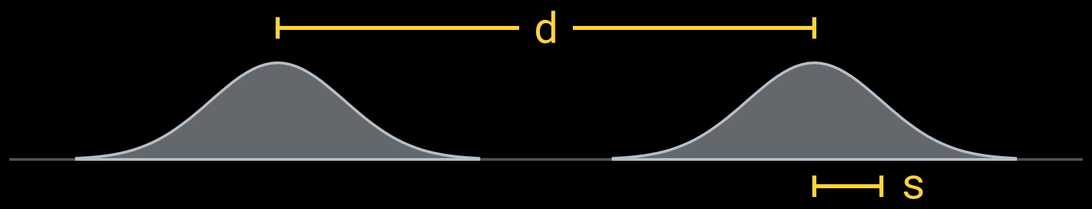
a longer window removes the errors, but it also cuts the symbols per second.

simulated readings; the trade-off is the one your own link shows

distance against spread

the distinguishability lab: hands-on-computer-science.de/cs/en/items/distinguishability-lab

add symbols to the alphabet until the first errors appear
lengthen the window until they are gone: what is left of the rate?
press disturb: which alphabet survives the stray light?



bigger d

- **pick colours further apart**
  fewer symbols fit into the range
- **use fewer symbols**
  fewer bits per symbol
- **aim the light better**
  only craft

smaller s

- **longer integration time**
  fewer symbols per second
- **average several readings**
  fewer symbols per second
- **block ambient light**
  only craft
- **fix distance and angle**
  only craft

three of them cost nothing but craft, which is why self-built optics are allowed.

part 3

build evidence

three ways to answer the same question

it sounds right

longer readings
collect more light

three ways to answer the same question

it sounds right

longer readings
collect more light

the assistant predicts it

```
double the
integration time
```

three ways to answer the same question

it sounds right

longer readings
collect more light



the assistant predicts it

`double the integration time`



the series shows it

`41 / 50 → 47 / 50 recognised correctly`



evidence from your setup

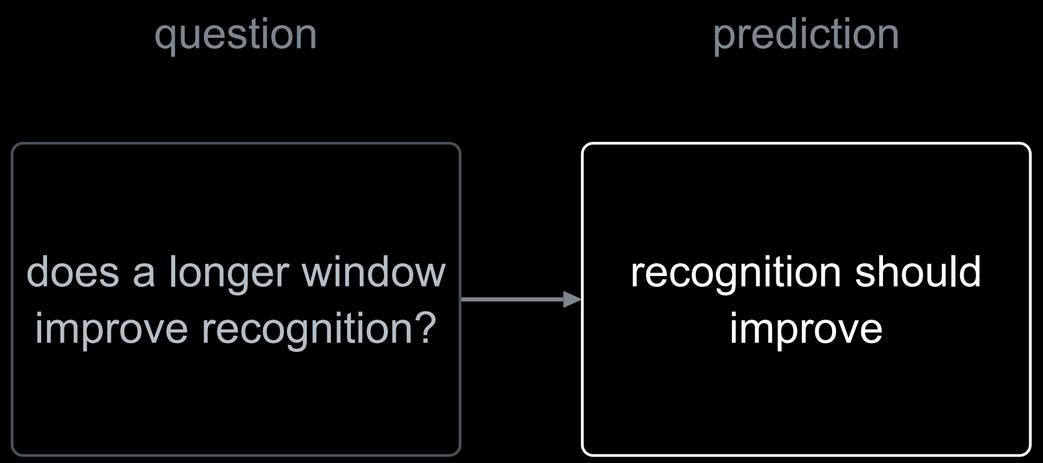
an explanation only claims something, while a measurement can decide it.

turn a claim into a test

question

does a longer window improve recognition?

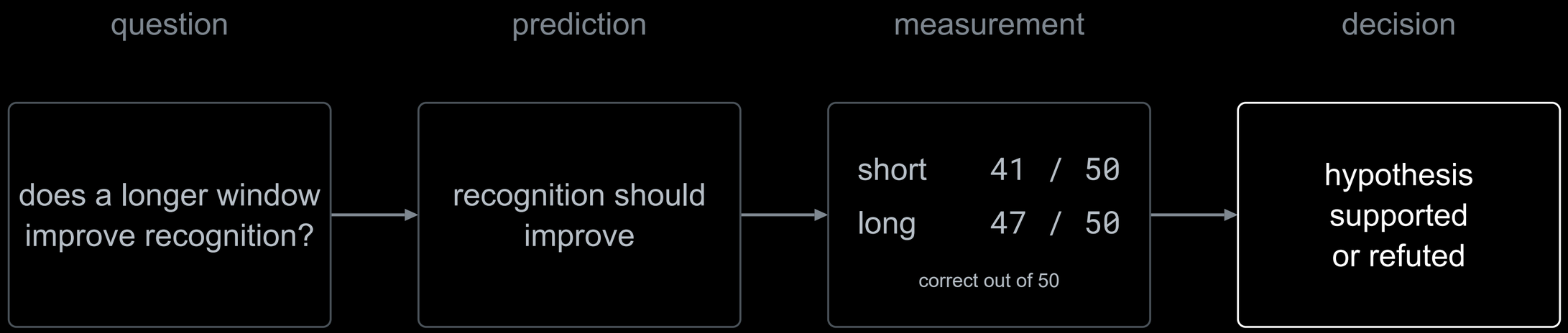
turn a claim into a test



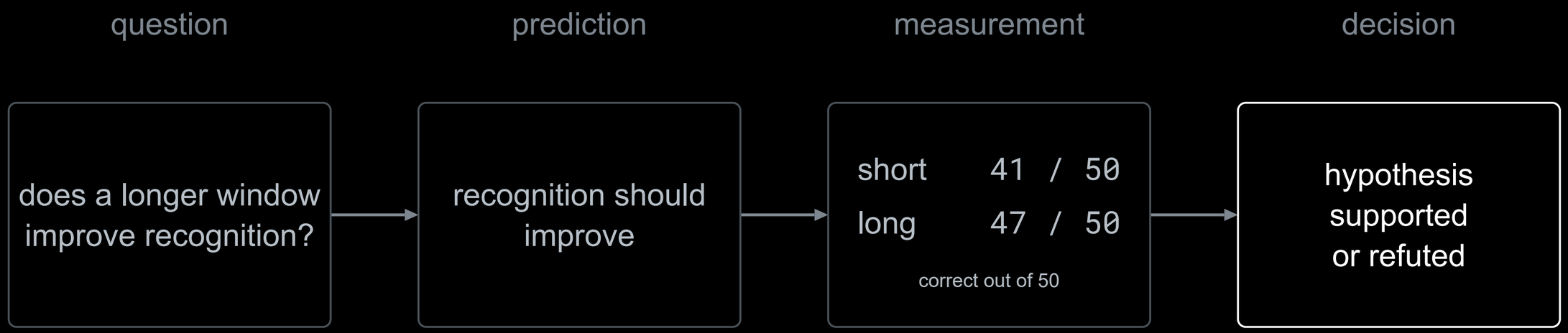
turn a claim into a test



turn a claim into a test



turn a claim into a test



write the expectation before you measure.

two teams improve their result

team a

changed distance and integration time

recognition improved

team b

changed only distance

recognition improved

two teams improve their result

team a

changed distance and integration time

recognition improved



what caused it?

team b

changed only distance

recognition improved



`distance`

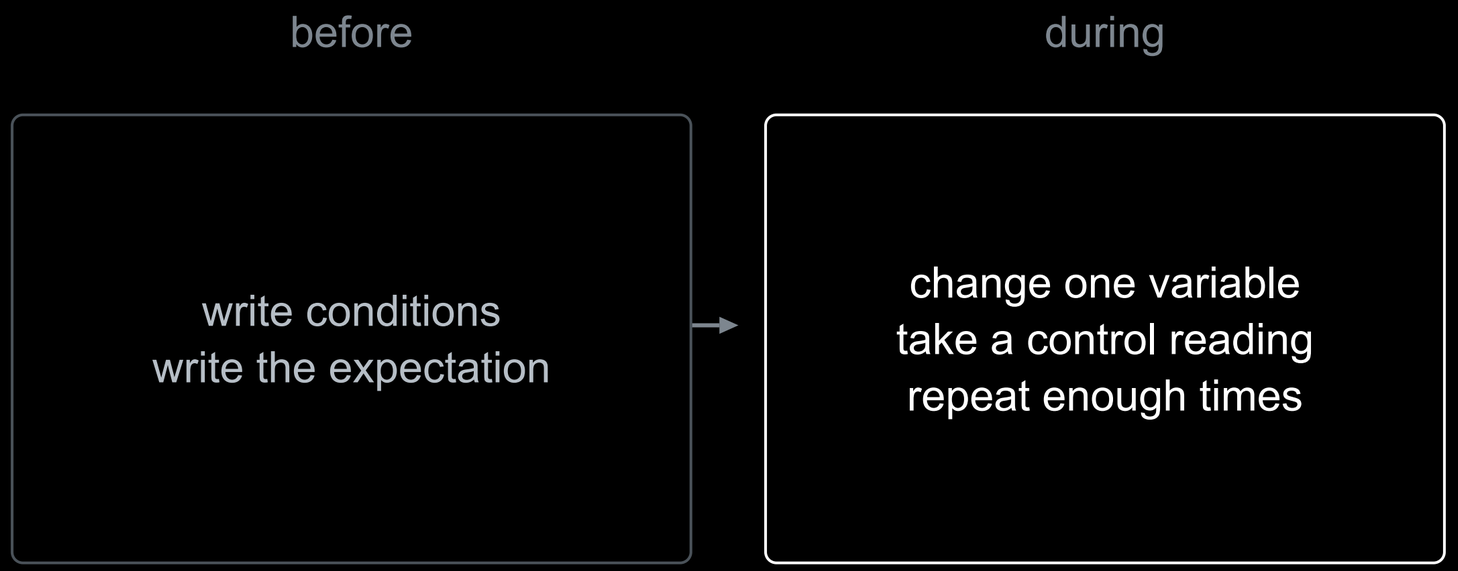
change one variable at a time.

what makes a measurement series

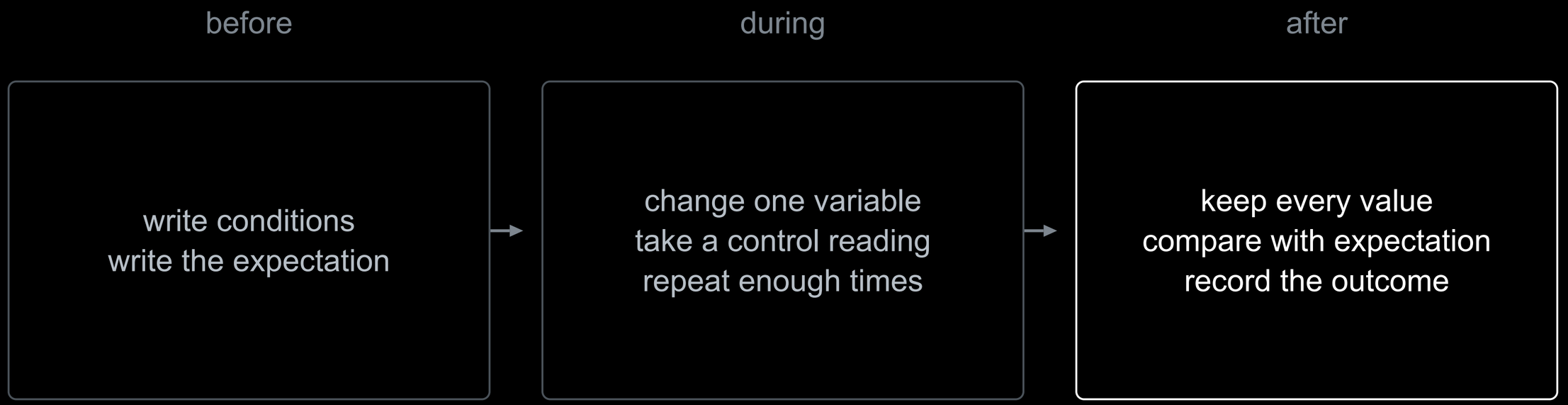
before

write conditions
write the expectation

what makes a measurement series



what makes a measurement series



a repeatable result is worth more than an impressive one.

a coin flip: luck or skill?

two colours: a guess is right half the time, like a coin flip



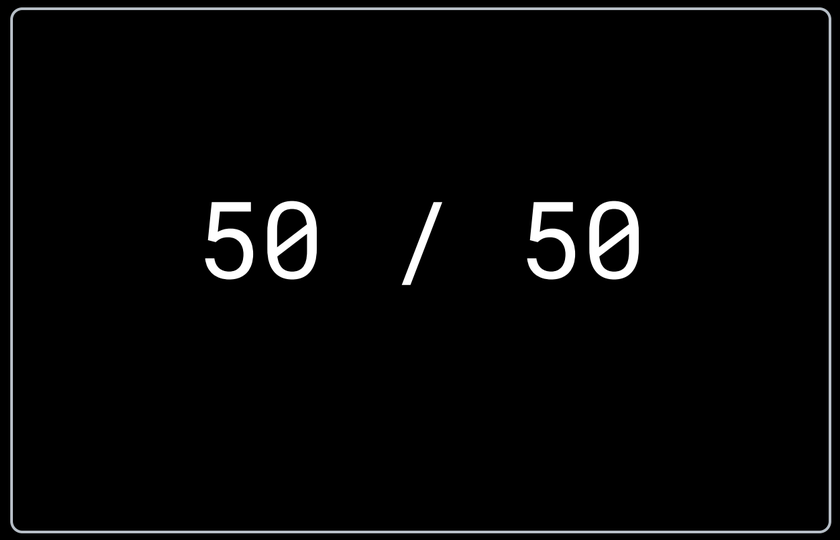


guessing succeeds about once in 32 attempts

two colours: a guess is right half the time, like a coin flip



guessing succeeds about once in 32 attempts



`guessing: about 1 in 1,125,899,906,842,624`

two colours: a guess is right half the time, like a coin flip



guessing succeeds about once in 32 attempts



`guessing: about 1 in 1,125,899,906,842,624`

evidence

when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"

your measurement series

```
gain ↑
recognition: 94% → 82%
```

when the assistant and the data disagree

assistant suggestion

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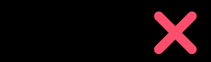
```
gain ↑
recognition: 94% → 82%
```

✓ one variable ✓ enough repetitions ✓ control reading

when the assistant and the data disagree

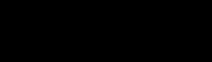
assistant suggestion

"increase the gain —
recognition will improve"



your measurement series

```
gain ↑
recognition: 94% → 82%
```



✓ one variable ✓ enough repetitions ✓ control reading

ground truth for your setup

the fog stays, so measure how far you can see.

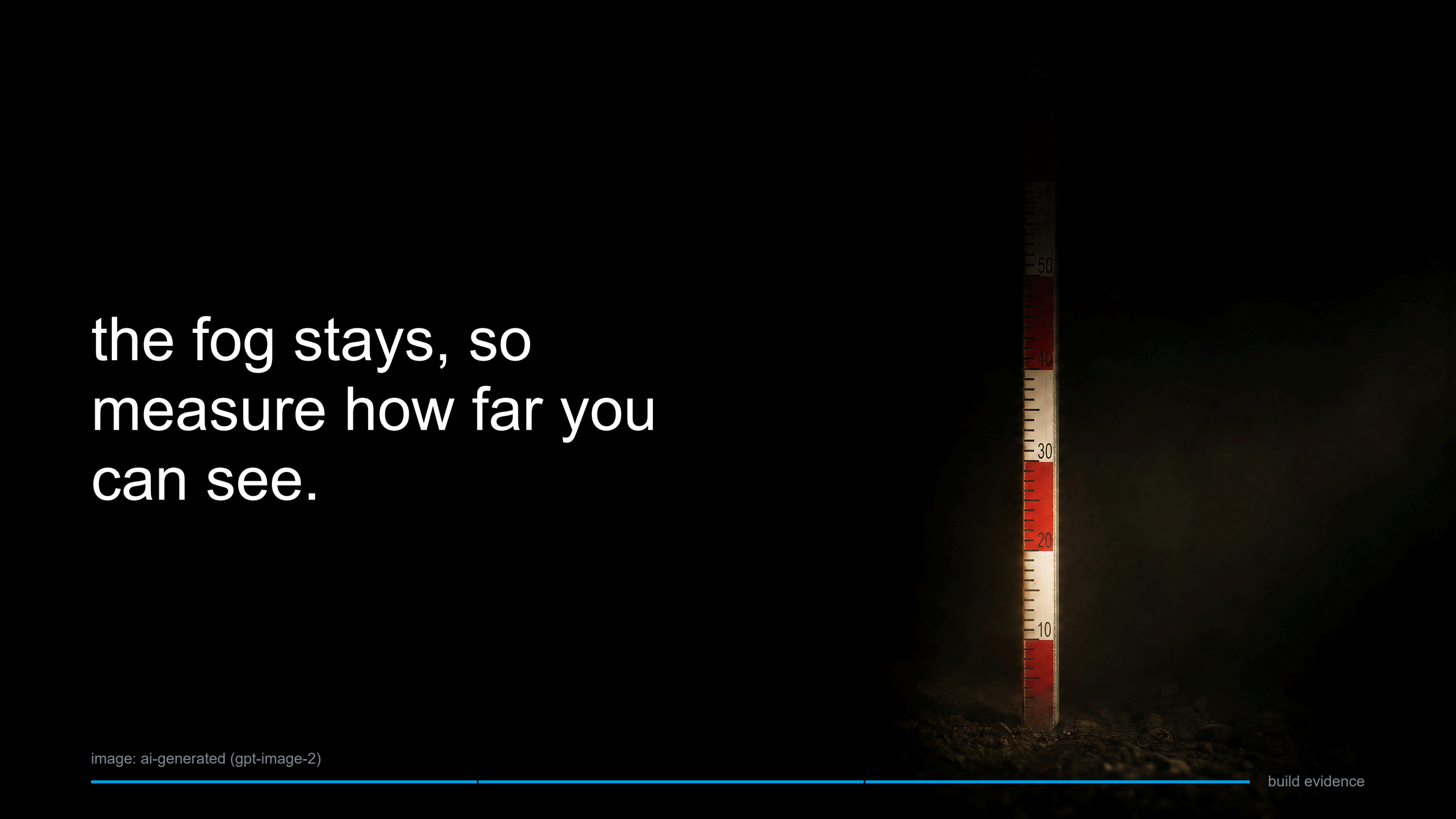


image: ai-generated (gpt-image-2)